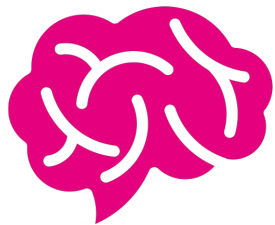


Dissolving

Lesson objectives

- To understand that the mass of a solution is mass of solute + mass of solvent.
- To demonstrate that a solute dissolves in a solvent to form a solution.
- To use particle theory to explain dissolving.



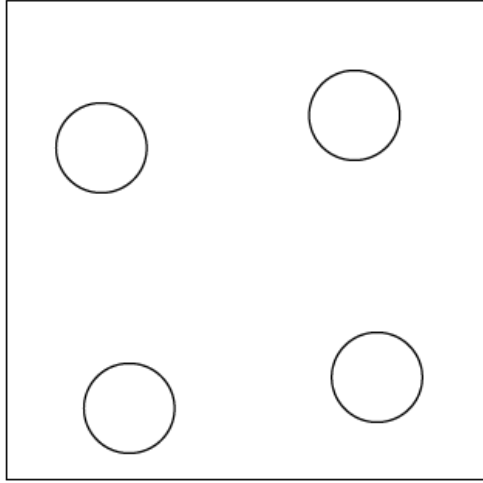
On your whiteboard, draw a diagram of the way particles are arranged in a **liquid**.

Keywords: solute, solvent, solution, dissolve, particles

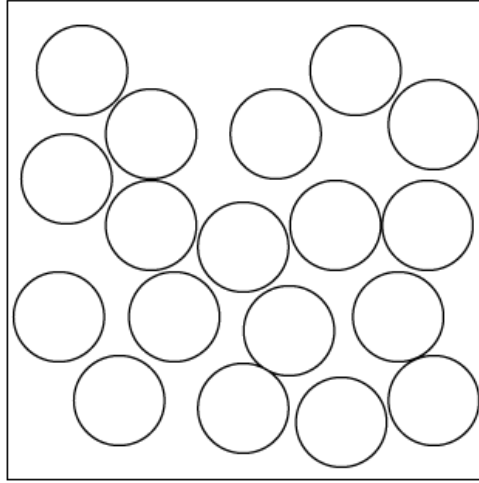
Success Criteria		
Working towards	Working at	Working beyond
I can identify a solute, solvent, and solution in a mixture.	I can measure mass/volume and describe dissolving using the particle model.	I can explain how particles interact in a solution and relate solute-solvent relationships

Dissolving

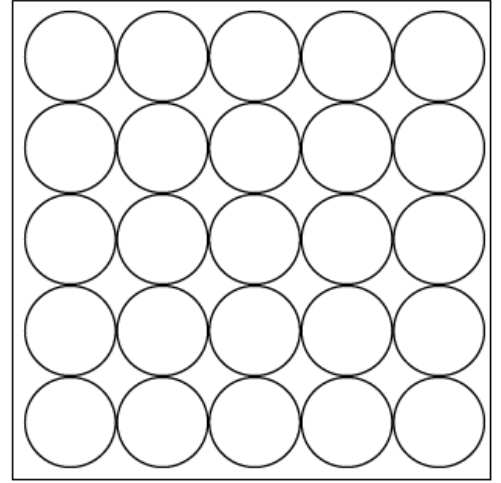
Solid, liquid and gas



X



Y

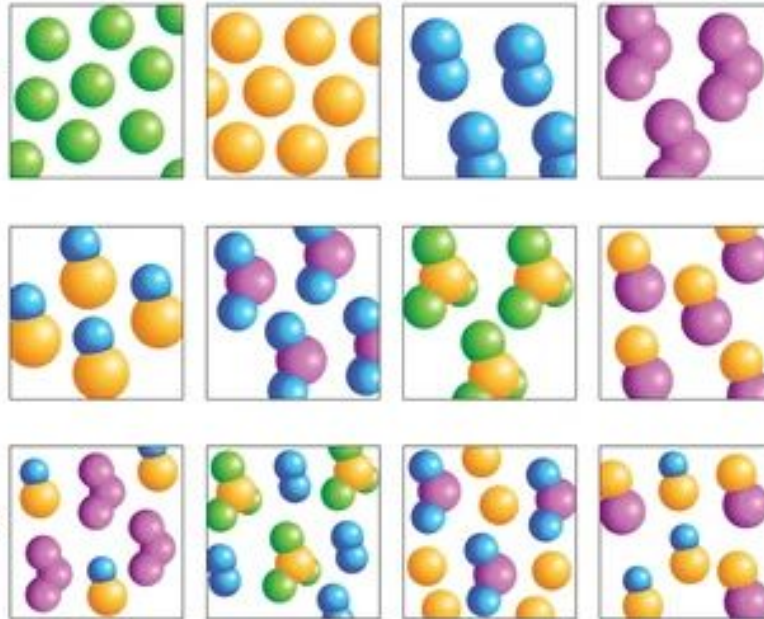


Z

What is X, Y and Z?

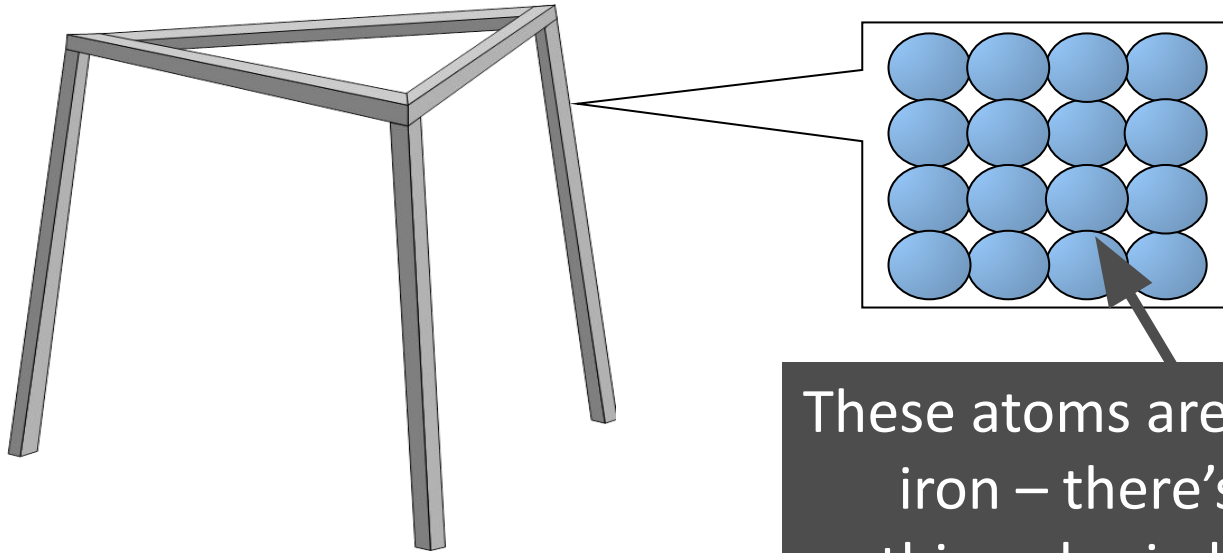
Elements, compounds and mixtures

What are the differences between an **element**, **compound** and **mixture**?



Element

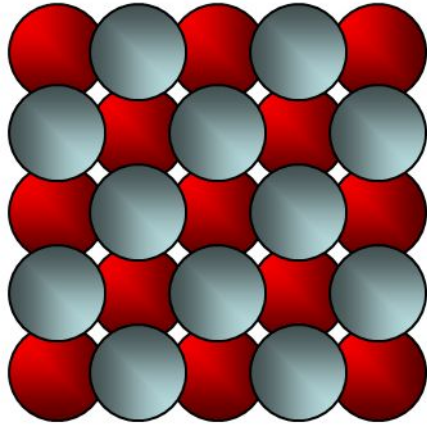
One type of atom, on its own or joined to another atom of the same type.



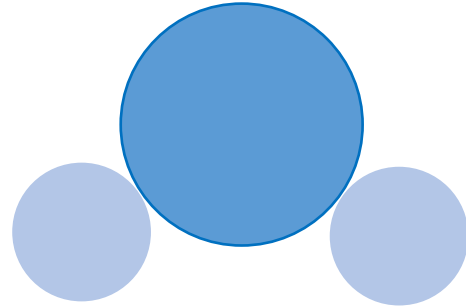
These atoms are ALL
iron – there's
nothing else in here

Compound

Different types of atom joined by a chemical bond.



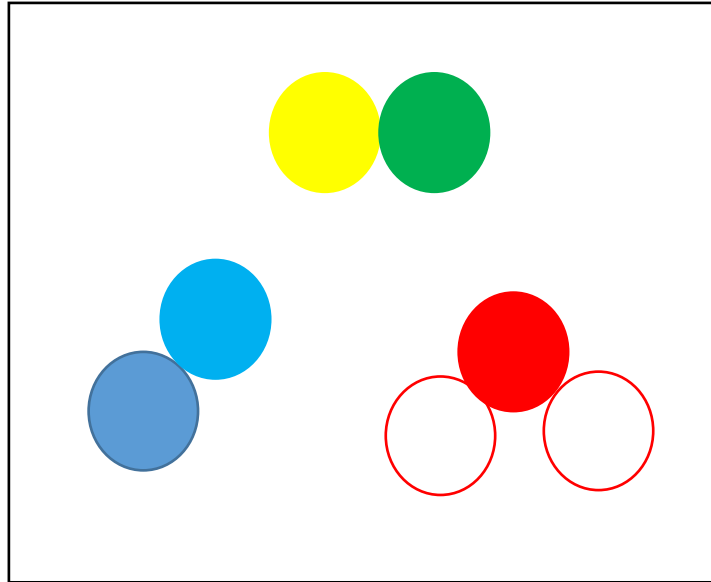
Sodium
chloride (salt)



Water
2 Hydrogen and 1
Oxygen

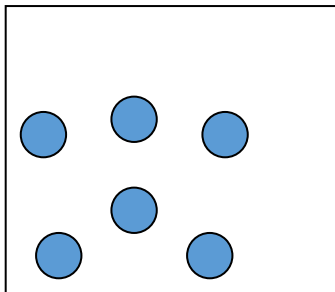
Mixture

Different types of atoms or compounds that can be physically separated,
NO chemical bond.

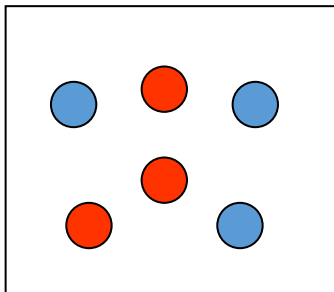


Element, compound, mixture

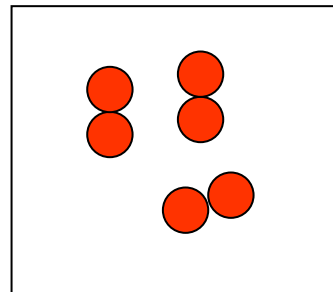
A



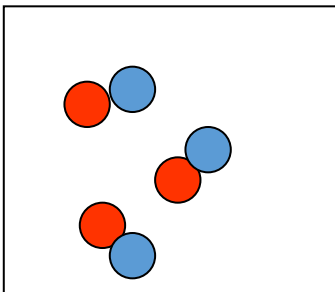
B



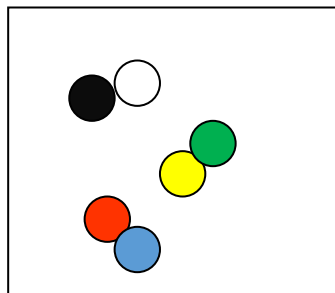
C



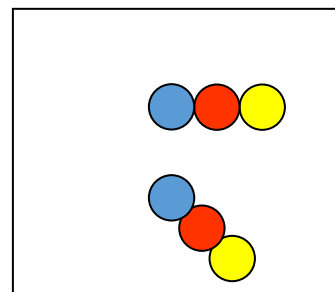
D



E

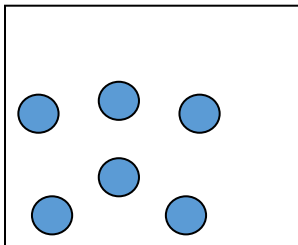


F



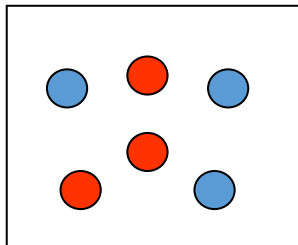
Element, compound, mixture

A



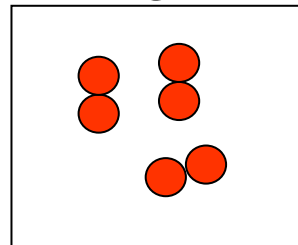
Element

B



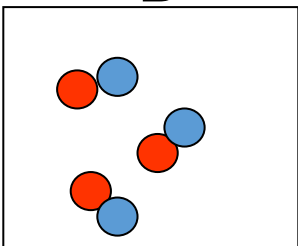
Mixture
(of elements)

C



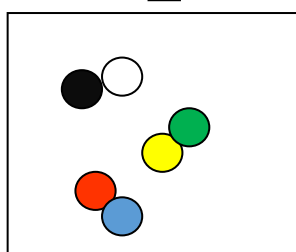
Element

D



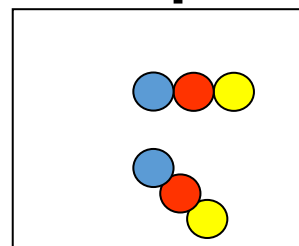
Compound

E



Mixture
(of compounds)

F



Compound

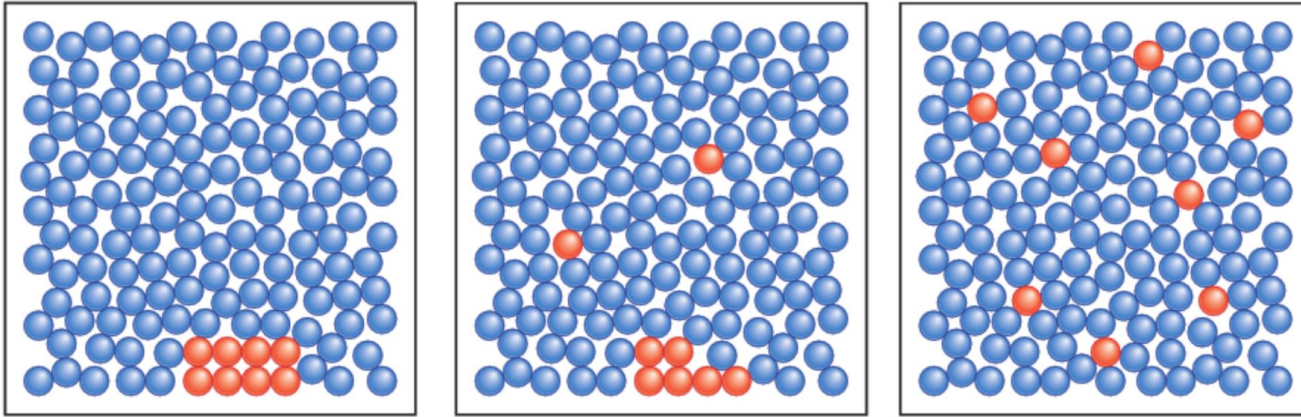


What happens to sugar when
it is put in water?

Dissolving

The sugar **DOES NOT** disappear!

It is still there but now separate particles spread around the water.



The diagram below shows what happens when particles dissolve.

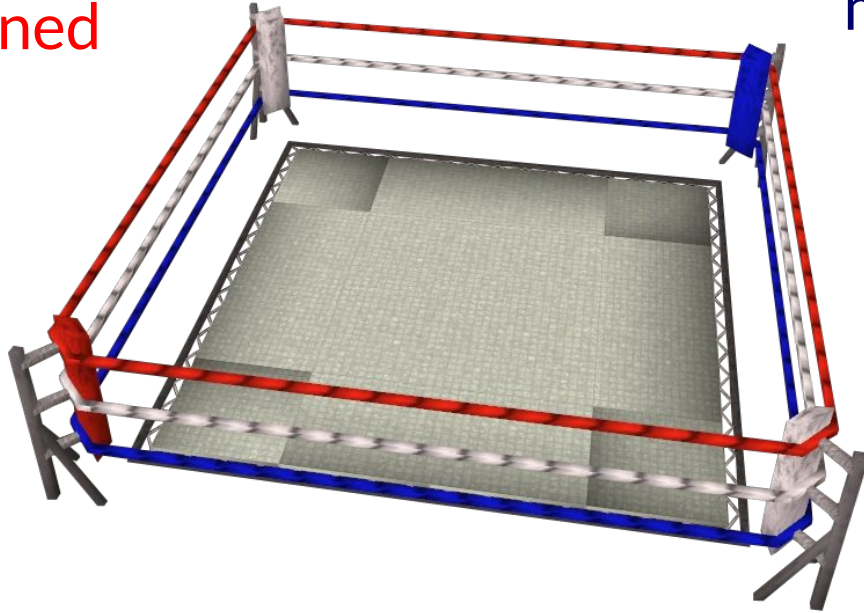
A substance that **can** dissolve is **soluble** in water.

A substance that **cannot** dissolve is **insoluble** in water.

Mixture vs Solution

two or more
substances that are
mixed together but
not chemically joined

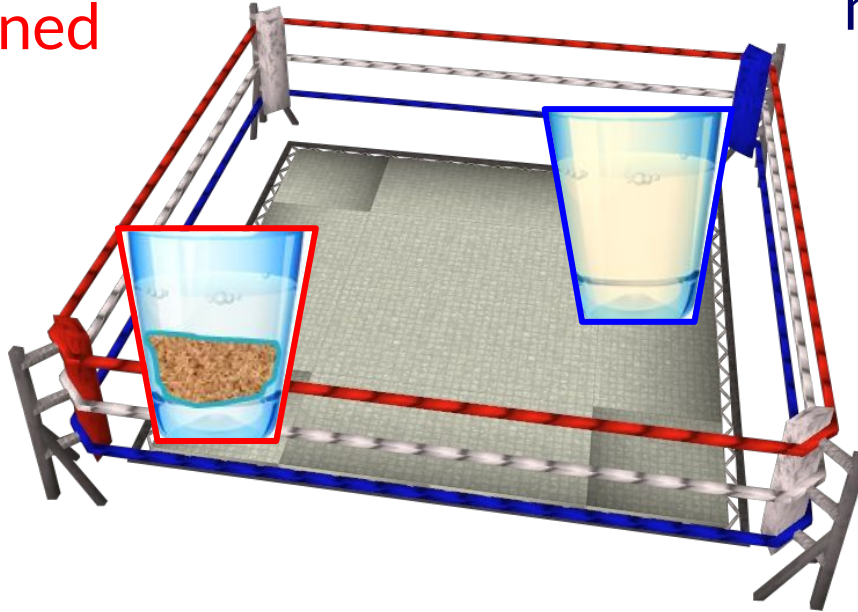
a special type of mixture
that is made when
a solid **dissolves** and
mixes a liquid.



Mixture vs Solution

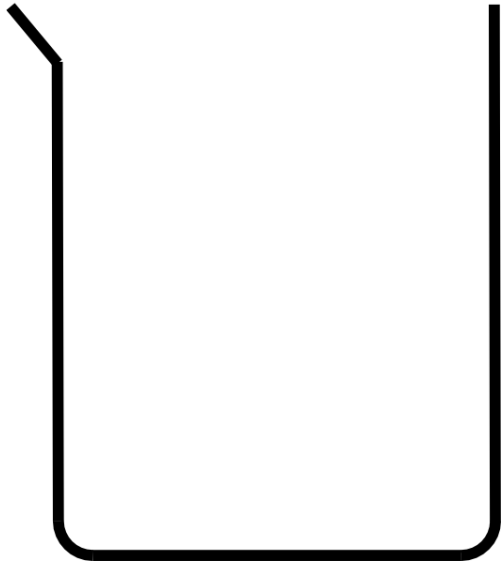
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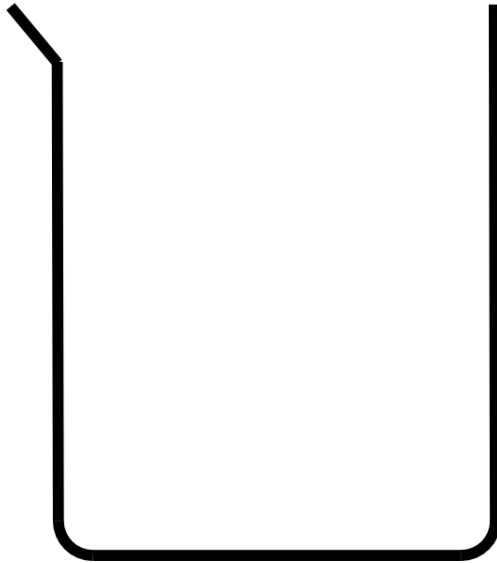


Dissolving

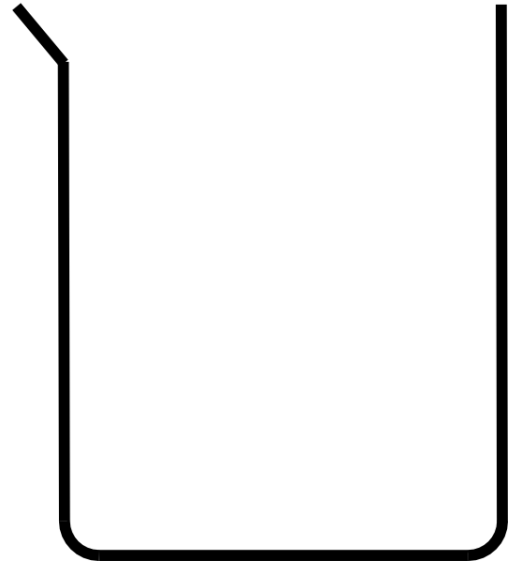
TASK: Use the particle model to explain dissolving



Particles in
solid sugar.



Particles in
liquid water.



Particles in
sugar solution.

solute + solvent = solution

solid being
dissolved



sugar

+

liquid that does
the dissolving



water

=

the dissolved
solute and
solvent



sugar water

How many other
examples of
solutions can
you think of?

Dissolving and mass

Think like a scientist

Dissolving and mass

You will need:

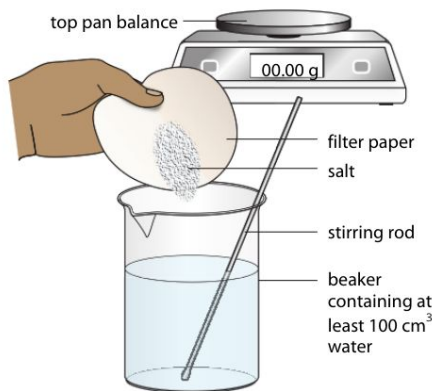
- top pan balance
- filter paper
- salt
- stirring rod
- beaker of water

Method

- 1 Place the filter paper on the top pan balance. Measure and record its mass.
- 2 Add about 20–25 g of salt. This is the solute. Measure and record the mass of the salt.
- 3 Remove the paper and salt from the balance.
- 4 Place a beaker containing at least 100 cm³ of water on the top pan balance. Measure and record the mass of the water and the beaker together. The water is the solvent.
- 5 Add the salt to the water. Stir until all the salt has dissolved. Measure the mass of the beaker and salt solution.

Questions

- 1 What was the mass of salt used?
- 2 What was the mass of the water and the beaker?
- 3 What was the mass of the solution and the beaker?
- 4 What does this tell you about the salt solution?



You are going to investigate how the mass of a solution is affected when a solute dissolves in a solvent.

1. Record the mass of your solvent (water)
2. Record the mass of your solute (salt)
3. Add the solute to the solvent and record your observations
4. Record the mass of your solution

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book page 43

Dissolving and Mass

What do you think will happen to the mass when we dissolve sugar in water?

- a) The sugar water will be lighter than the water and sugar separately.
- b) The mass of the sugar water will be equal to the mass of the sugar and water separately.
- c) The sugar water will be heavier than the sugar and water separately.

solute + **solvent** = **solution**

solid being
dissolved

liquid that does
the dissolving

the dissolved
solute and the
solvent



sugar

9g

+



water

146g

=



sugar water

155g

solute + solvent = solution

solid being
dissolved



sugar

9g

+

liquid that does
the dissolving



water

146g

=

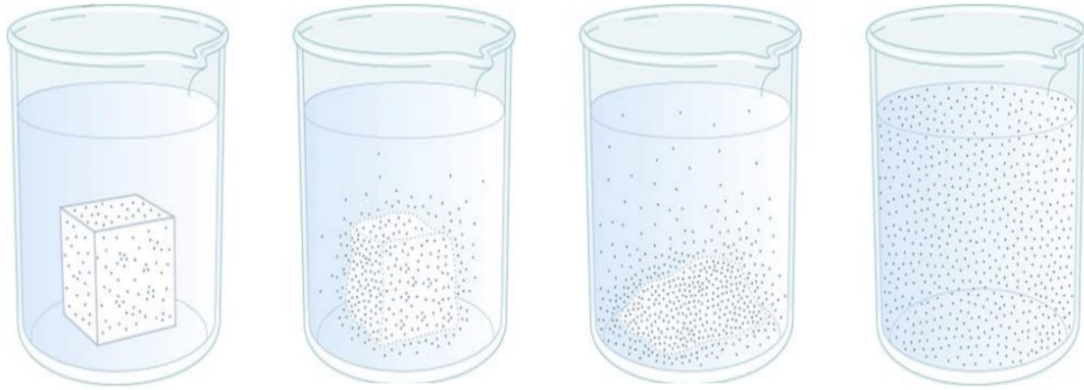
What would happen to the mass of sugar if this solution was made

- a) more concentrated
- b) more dilute

155g

**The mass of solute and solvent equals mass of the solution.
No mass has been lost. The mass has been conserved.**

Dissolving and melting



2

1

Plenary

- 1 In a solution of sugar and water, which is the solvent and which is the solute?
- 2 What is the difference between dissolving and melting?
- 3 What mass of salt solution is made when 9 g of salt is dissolved in 50 g of water? Explain how you worked out your answer.
- 4 A green powder was placed into a beaker of water. After it was stirred, the water looked cloudy and lumps of powder could still be seen. Has a solution been formed? Explain your answer.

Plenary

- 1 Water is the solvent; sugar is the solute.
- 2 When something dissolves, such as sugar in tea, there are two substances involved. The sugar is the solute, which dissolves in the tea, which is the solvent. When something melts, such as ice cream on a hot day, only one substance is involved.
- 3 59 g; the mass of water is 50 g and the mass of the salt is 9 g so together it is 59 g.
- 4 No; a solution is transparent and if there are lumps and it is cloudy, it is not a solution. The green powder was insoluble.